

Do we co-contract in order to learn?

Preliminary simulation evidence for H0 (URF proposal *Tuning to Learn*)

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September 3, 2026

The claim (H0)

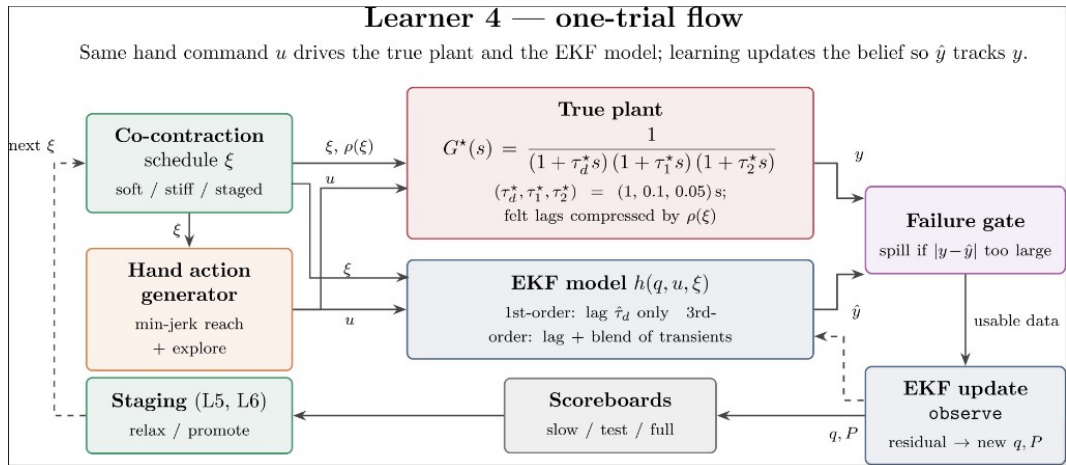
A complicated object is made **learnable** by tuning your own body to **simplify** it — not only to stabilise it.

Question tested here

Does bracing actually make the dominant dynamics faster and safer to learn — and can you still recover the rest later?

- Carry a cup of coffee: a **slow** displacement (dominant dynamics) plus a **fast** wobble (transients).
- Co-contraction (bracing) pushes the fast poles deeper into the left half-plane.
- The fast wobble settles *within the movement* $\Rightarrow$ it becomes filtered noise.
- What is left to learn is only the dominant, slow dynamics.

Overall block diagram



What is a trial? What is a seed?

One trial

One out-and-back reach with the cup.

- Duration $T = 2.5$ s, sample $\Delta t = 0.01$ s ($n_t = 250$ samples).
- Movement time fixed: $t_{\text{move}} = 0.25$ s.
- Pick grip ξ , generate hand path u , feel y , predict $\hat{y}$, maybe update belief.
- Can **fail** (“spill”) if $|y - \hat{y}|$ exceeds tolerance — then learning uses only the usable prefix, or none.

Figures report medians over **60 trials** per seed.

One seed

One independent run of the whole experiment (a fresh random generator).

- Exploration noise and sensor noise differ across seeds.
- Same strategy, same plant, different random realisations.
- We run **20 seeds** and report medians / means so one lucky noise draw cannot decide the story.

Think: 20 simulated people, each doing 60 reaches under one strategy.

Block diagram — notation and formulas

Symbol	Meaning / formula
ξ	Co-contraction (grip). Soft $\xi \approx 0.7$; stiff $\xi \approx 3$.
$\rho(\xi)$	Separation ratio: $\rho = (\xi + \sqrt{\xi^2 - 1})^2$ if $\xi > 1$, else 1.
u	Hand command: $u = u_{\text{ref}} + 0.10 u_{\text{explore}}$ (min-jerk + band-limited noise).
$G^*(s)$	True plant: $\frac{1}{(1 + \tau_d^* s)(1 + \tau_1^* s)(1 + \tau_2^* s)}$, $\tau^* = (1, 0.1, 0.05) \text{ s}$.
Felt lags	$(\tau_d^*, \tau_1^*/\rho, \tau_2^*/\rho)$ — only fast poles compressed.
y	Measurement: cascade of felt lags on u , plus noise (0.012ε).
q, P	EKF belief and covariance (1st-order: $\log \hat{\tau}_d$; 3rd: also $\log \hat{\tau}_{1,2}, w_{1,2}$).
$h(q, u, \xi)$	Measurement model $\rightarrow$ prediction $\hat{y}$.
n_{eff}	Effective order = $1 + w_1 + w_2$ (how many modes are “on”).
Failure	Spill if $ y[k] - \hat{y}[k] > 0.15 \cdot \xi / \xi_{\text{soft}}$.

One-trial flow (through the diagram)

1. **Schedule** picks ξ for this trial (soft / stiff / staged).
2. **Hand action generator** builds u (same out-and-back path + small wiggles).
3. **True plant** gets (u, ξ) : compresses fast lags by $\rho(\xi)$, cascades them, adds noise $\rightarrow$ felt y .
4. **EKF model** gets the *same* (u, ξ) and current belief $q \rightarrow$ predicted feel $\hat{y} = h(q, u, \xi)$.
5. **Failure gate** compares y and $\hat{y}$. If mismatch is too large, truncate (spill); else keep the trial.
6. **EKF update** (observe): residual $y - \hat{y}$ nudges q and shrinks P (or drift if too little usable data).
7. **Scoreboards** log slow / soft-probe / full-object error [how well we know the slow lag; whether we could carry the cup softly now; and how good our full model of the cup is].
8. **S3 (brace then hand over)**: if confident enough, relax ξ and hand the slow estimate to a fuller soft model (S2 stays braced forever and never does this step).

Key point

Same u drives world and brain. Learning is nothing more than updating q so that $\hat{y}$ tracks y .

The object and the learner

True plant (never changes):

$$G(s) = \frac{1}{(1 + \tau_d s)(1 + \tau_1 s)(1 + \tau_2 s)}$$

$$(\tau_d, \tau_1, \tau_2) = (1.0, 0.1, 0.05) \text{ s}$$

One slow pole (dominant dynamics), two fast poles (wobbles).

Co-contraction ξ compresses the fast poles by

$$\rho(\xi) = (\xi + \sqrt{\xi^2 - 1})^2, \quad \xi > 1.$$

The learner: an extended Kalman filter (EKF)

- Carries a belief over the time constants.
- Predicts the felt force from hand motion + grip.
- Updates from the residual (predicted vs. felt).
- Unused fast modes are pruned toward “off” (automatic relevance prior).
- Effective model order n_{eff} reads off how many modes currently participate.

How $G(s)$ is scaled by $\rho(\xi)$

Object itself never changes:

$$G^*(s) = \frac{1}{(1 + \tau_d^* s)(1 + \tau_1^* s)(1 + \tau_2^* s)}, \quad (\tau_d^*, \tau_1^*, \tau_2^*) = (1.0, 0.1, 0.05) \text{ s.}$$

What co-contraction changes is the *felt* plant: only the fast lags are divided by $\rho(\xi)$:

$$\rho(\xi) = \begin{cases} 1 & \xi \leq 1, \\ (\xi + \sqrt{\xi^2 - 1})^2 & \xi > 1, \end{cases} \quad G_{\text{felt}}(s) = \frac{1}{(1 + \tau_d^* s)(1 + \frac{\tau_1^*}{\rho} s)(1 + \frac{\tau_2^*}{\rho} s)}.$$

Grip	ξ	$\rho(\xi)$	Felt lags ($\tau_d, \tau_1/\rho, \tau_2/\rho$)
Soft	0.7	1	(1.0, 0.10, 0.05) s — all three visible
Stiff	3.0	≈ 34	(1.0, 0.0029, 0.0015) s — wobbles almost gone

In plain words

Bracing does not change the cup. It makes the fast wobbles finish so quickly that, during one reach, you effectively feel only the slow lag.

Strategies compared

Same movement speed throughout — only co-contraction and model capacity vary.

S1 — Deep end

Low co-contraction from trial 1, full three-mode model.

S2 — Brace and keep

High co-contraction throughout, learning only the dominant dynamics.

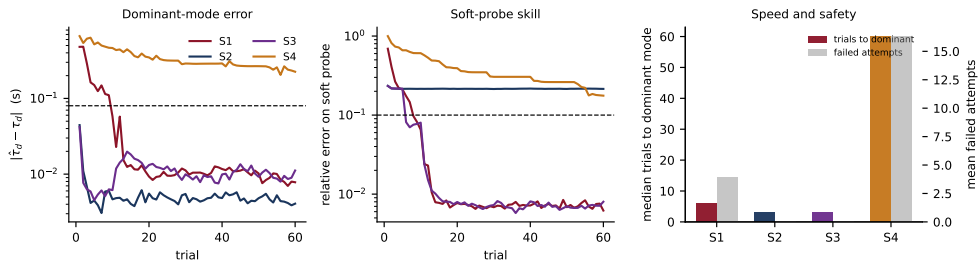
S3 — Brace then hand over

High co-contraction first, then relaxed in stages into a full low-co-contraction model.

S4 — Soft, slow-only (control)

Low co-contraction like S1, but tries to learn only the dominant dynamics: tests whether a simpler model helps *without* bracing.

Results



20 seeds, 60 trials. Left: dominant-dynamics error. Middle: common soft-probe skill. Right: trials & failures vs. S1.

Reading the results — left and right panels

Left: dominant-mode error

What it plots. $|\hat{\tau}_d - \tau_d|$ — how wrong is the learner's estimate of the **slow** lag (the main “heavy” response of the cup).

In plain words. “Do I know the slow part of the object yet?” Lower = better.

What you see.

- S2/S3 drop fast and stay low (bracing filters the wobble).
- S1 gets there too, but more slowly and with a big early swing.
- S4 never arrives: without bracing, the fast wobble systematically biases $\hat{\tau}_d$.

Right: speed and safety

What it plots. Coloured bars = median trials to get the slow lag right ($|\hat{\tau}_d - \tau_d| < 0.08$ s). Grey bars = mean failed (“spill”) attempts.

In plain words. “How many reaches until I know the slow part?” and “How often did I drop the cup while learning?”

What you see. S2 and S3 are **faster** (3 vs 6 trials) and **safer** (0 spills vs ~ 4 for S1). S4 never finishes and spills most.

Reading the results — soft-probe skill (middle)

What is the soft probe?

After every training trial we ask a **fixed exam question** that is the same for every strategy:

- a fast out-and-back reach,
- always at **low** co-contraction ($\xi \approx 0.7$),
- so **all three poles are visible** (slow + both wobbles).

The curve is the relative error of the current belief predicting that soft reach. Criterion: error < 0.1 .

What is it used for?

Training can happen while braced. Soft-probe skill asks: *“If I had to carry the cup softly now, would my model still work?”* It stops a braced learner from looking “done” when it only knows the slow lag and has never seen the wobbles. That is why S2 (always stiff) looks great on the left panel but stays flat and high in the middle (~ 0.22) — it never learned the rest.

Results — numbers

	trials to $ \hat{\tau}_d - \tau_d < 0.08$ s	failed attempts	peak error
S1 deep end	6	~ 4	~ 0.69 s
S2/S3 braced	3 (−50%)	0 (−100%)	~ 0.02 s (−97%)
S4 soft, slow-only	never	~ 16 (27%)	plateaus +0.22 s bias

- Braced strategies (S2, S3) are **faster** and **safer** on the dominant dynamics.
- S4 shows the filter, not just a smaller model, is doing the work: without bracing the fast wobbles corrupt $\hat{\tau}_d$.
- On the common **soft probe** (all 3 poles visible): S2 never passes (~ 0.22 plateau); S3 passes with **zero failures**, matching S1's soft accuracy after hand-over.

- **Co-contraction is not a crutch you discard** — it buys a faster, safer route to the part of the task that matters first.
- **High co-contraction forever is not the answer**: S2 wins the dominant dynamics but never learns the rest.
- **Staged release is the missing piece**: S3 hands over the converged slow estimate and recovers full soft skill at the cost of only a few extra trials.
- The plant does the simplifying, not a hand-coded restriction: a full-capacity learner held stiff prunes *itself* to $n_{\text{eff}} = 1$.

Takeaway for H0

Co-contraction is a self-authored curriculum over model complexity: brace to learn the dominant dynamics fast and safely, then relax in stages to recruit the rest. This is the preliminary mechanism WP1 tests in people.